Question Bank: PostgreSQL Query Performance & Indexing

QUESTIONS.MD | CANDIDATE | 2026-09-02

Candidate copy. Assessment material has been removed from this document: 48 field(s) from the source were withheld. Only the task statements and the logistics needed to attempt them are included.

Q-L1-01

THEORY · WEIGHT 2

> ASK A B-tree index exists on `orders(created_at)`. Which of these can it serve, and which can't it: a range filter on `created_at`, a sort by `created_at`, and a filter on `status`? Say why in each case.

TIME BUDGET 3 min

Q-L1-02

THEORY · WEIGHT 2

> **ASK** You have an index on `orders(customer_id, created_at)`. A query filters only on `created_at`. Will the index be used? Explain.

TIME BUDGET 3 min

Q-L2-01

DEBUG · WEIGHT 3

> ASK There's a B-tree index on `orders(created_at)`. This query is doing a sequential scan over 80 million rows: `SELECT * FROM orders WHERE date(created_at) = '2026-01-15'`. The index is valid and the table was analyzed this morning. Why isn't the index used, and what would you change?

TIME BUDGET 5 min

Q-L2-02

PRACTICAL · WEIGHT 3

> ASK Here's a line from EXPLAIN ANALYZE on a query blowing the 200ms budget: `Index Scan using idx_orders_created_at on orders (cost=0.43..8.45 rows=1 width=248) (actual time=0.02..412 rows=2100000 loops=1)`. Tell me what this line says, and where you'd look next.

TIME BUDGET 5 min

Q-L3-01

TRADEOFF · WEIGHT 4

> **ASK** The orders table takes about 8,000 inserts per second at peak. A team wants to add three new indexes to speed up their reporting queries. You have to give an answer. What is it, and what does it cost you?

TIME BUDGET 6 min

Q-LC-01

LIVECODE · WEIGHT 5

> **ASK** Write a single SQL query against `orders(id, customer_id, status, total_cents, created_at)` — about 200 million rows, with a B-tree index on `(customer_id, created_at)`. For every customer who ordered in the last 30 days, return their most recent order in that window, together with how many orders they placed in that window. It has to run on the primary inside the 200ms budget. Tell me when you're done, and I'll read it.

TIME BUDGET 20 min

MODE review-only

RUNTIME PostgreSQL 16 — no live database in this session; the query is scored by reading it. (A `runnable` kit would supply a seeded database and run the hidden tests at submission.)

STARTER FILE `livecode/Q-LC-01/solution.sql`

STARTER CONTENTS

```
-- orders(id bigint, customer_id bigint, status text, total_cents int, created_at timestamptz)
-- index: orders_customer_created_idx ON orders (customer_id, created_at)
-- Write your query below.
```